Woojung Han

Website | Google Scholar | LinkedIn | GitHub

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Summary

Driven by a strong passion for learning new things, I am broadly interested in algorithms that bridge **visual understanding** and **multi-modality**. My core focus lies in multimodal understanding and generation, including vision foundation models and image/video generation. Furthermore, I am actively expanding my research interests into emerging areas such as World Models, MLLMs.

Education

Ph.D. in Computer Science — Seoul, South Korea
Yonsei University — *Mar 2022 – Present*
Advisor: Seong Jae Hwang

B.S. in Human Intelligence and Information Engineering — Seoul, South Korea
Sangmyung University — *Mar 2017 – Feb 2022*
Advisor: Eui Chul Lee

Experiences

Intern: Google Student Researcher — July 2026 – Present
Google Deepmind (Advisor: Li Zhang) — *Mountain View, CA*

- Efficient Video Representation Learning

Lab Intern: Algorithm Machine Intelligence Lab — Jun 2021 – Aug 2021
POSTECH (Advisor: Tae-Hyun Oh) — *Remote – Pohang, South Korea*

- Efficient Human Behavior Understanding
- Human Mesh Recovery

Lab Intern: Computer Vision Lab — Mar 2021 – Jun 2021
Korea University (Advisor: Sangpil Kim) — *Seoul, South Korea*

- Hand Grasp Type Estimation

Lab Intern: Pattern Recognition Lab — Sep 2018 – Nov 2020
Sanmyung University (Advisor: Eui Chul Lee) — *Seoul, South Korea*

- Real-time Remote Photoplethysmography Extraction

+ Attended Google BootCamp in 2022

Publications (* Equal Contribution)

Conference Publications

1. *Physics in 2-Steps: Locking Motion Priors Before Visual Refinement Erases Them*
   Woojung Han, Seil Kang, Youngjun Jun, Min-Hung Chen, Fu-En Yang, Seong Jae Hwang
   Forty-Third International Conference on Machine Learning (**ICML**), 2026 [paper]

2. *Real-Time Visual Attribution Streaming in Thinking Model*
   Seil Kang, Woojung Han, Junhyeok Kim, Jinyeong Kim, Youngeun Kim, Seong Jae Hwang
   Forty-Third International Conference on Machine Learning (**ICML**), 2026 [paper]
   Spotlight (Acceptance Rate < 2.2%)

3. *Interpretable Motion-Attentive Maps: Spatio-Temporally Localizing Concepts in Video Diffusion Transformers*
   Youngjun jun, Seil Kang, Woojung Han, Seong Jae Hwang
   Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026 [paper]
   Highlight(Acceptance Rate < 3.6%)

4. *Rare Text Semantics Were Always There in Your Diffusion Transformer*
   Woojung Han*, Seil Kang*, Dayun Ju, Seong Jae Hwang
   Neural Information Processing System (**NeurIPS**), 2025 [paper]

5. *PRETI: Patient-Aware Retinal Foundation Model via Metadata-Guided Representation Learning*
   Yeonkyung Lee, Woojung Han, Youngjun Jun, Hyeonmin Kim, Junkyung Cho, Seong Jae Hwang
   Conference on Medical Image Computing and Computer-Assisted Intervention (**MICCAI**), 2025 [paper]
   Early Accepted (Acceptance Rate <9%)

6. *Pathology-Aware Adaptive Watermarking for Text-Driven Medical Image Synthesis*
   Chanyoung Kim*, Dayun Ju*, Jinyeong Kim, Woojung Han, Roberto Alcover-Couso, Seong Jae Hwang
   Conference on Medical Image Computing and Computer-Assisted Intervention (**MICCAI**), 2025 [paper]
   Early Accepted (Acceptance Rate <9%)

7. *Spatial Transport Optimization by Repositioning Attention Map for Training-Free Text-to-Image Synthesis*
   Woojung Han, Yeonkyung Lee, Chanyoung Kim, Kwanghyun Park, Seong Jae Hwang
   Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025 [paper/page]

8. *Distilling Spectral Graph for Object-Context Aware Open-Vocabulary Semantic Segmentation*
   Chanyoung Kim, Dayun Ju, Woojung Han, Ming-Hsuan Yang, Seong Jae Hwang
   Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025 [paper/page]

9. *Advancing Text-Driven Chest X-Ray Generation with Policy-Based Reinforcement Learning*
   Woojung Han*, Chanyoung Kim*, Dayun Ju, Yumin Shim, Seong Jae Hwang
   Conference on Medical Image Computing and Computer-Assisted Intervention (**MICCAI**), 2024 [paper/project]
   Spotlight (Acceptance Rate <3.4%)

10. *EAGLE: Eigen Aggregation Learning for Object-Centric Unsupervised Semantic Segmentation*
    Woojung Han*, Chanyoung Kim*, Dayun Ju, Seong Jae Hwang
    Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024 [paper/ page]
    Highlight (Acceptance Rate <2.8%)

11. *Effect of Time Window Size for Converting Frequency Domain in Real-Time Remote Photoplethysmography Extraction*
    Yu Jin Shin, Woojung Han, Kun Ha Suh & Eui Chul Lee
    Conference on Intelligent Human Computer Interaction (**IHCI**), 2021 [paper]

Journal Publications

1. *Complementary Branch Fusing Class and Semantic Knowledge for Robust Weakly Supervised Segmentation*
   Woojung Han, Seil Kang, Kyobin Choo, Seong Jae Hwang
   Pattern Recognition(**PR**) [Impact Factor: 7.5], 2024 [paper]

Technical Reports

1. *When FLOPs Lie: Hardware-Faithful Token Merging for Diffusion Transformers on Mobile Devices*
   Hyunkyung Han, Woojung Han, Sujung Hong, Seil Kang, Seong Jae Hwang

2. *Reference Bias in Motion Transfer: Adapting Attention Queries for Target-Preserving Generation*
   Sunyoung Jung, Jiwoo Park, Yeonkyung Lee, Woojung Han, Seong Jae Hwang

Workshop Papers

1. *Physics in 2-Steps: Locking Motion Priors Before Visual Refinement Erases Them*
   Woojung Han, Seil Kang, Youngjun Jun, Min-Hung Chen, Fu-En Yang, Seong Jae Hwang
   Foundations of Deep Generative Models @ ICML 2026(**ICML Workshop**), 2026

2. *Real-Time Visual Attribution Streaming in Thinking Model*
   Seil Kang, Woojung Han, Junhyeok Kim, Jinyeong Kim, Youngeun Kim, Seong Jae Hwang
   Scalable Learning and Optimization for Efficient Multimodal AI Agents (SCALE) @ ICML 2026 Best Paper Award(**ICML Workshop**), 2026

3. *Spatial Transport Optimization by Repositioning Attention Map for Training-Free Text-to-Image Synthesis*
   Woojung Han, Yeonkyung Lee, Chanyoung Kim, Kwanghyun Park, Seong Jae Hwang
   3rd Workshop on Generative Models for Computer Vision @ CVPR 2025(**CVPR Workshop**), 2025 [paper]

4. *Distilling Spectral Graph for Object-Context Aware Open-Vocabulary Semantic Segmentation*
   Chanyoung Kim, Dayun Ju, Woojung Han, Ming-Hsuan Yang, Seong Jae Hwang
   The 4th Workshop on Transformers for Vision @ CVPR 2025(**CVPR Workshop**), 2025 [paper]

5. *EAGLE: Eigen Aggregation Learning for Object-Centric Unsupervised Semantic Segmentation*
   Woojung Han*, Chanyoung Kim*, Dayun Ju, Seong Jae Hwang
   Causal and Object-Centric Representations for Robotics @ CVPR 2024(**CVPR Workshop**), 2024 [paper]

Professional Services

**** Teaching Experience***

Samsung Korea *Teaching DX & AI Project* *2022*
Samsung Korea *Teaching AI & DX* *2022*
NAVER + MOBULLABS *Coach Machine Learning and Deep Learning (AI)* *2020, 2021, 2022*
KT AIVLE *Coach of AI* *2021*
LG Academy AI Camp *Coach of AI* *2021*

**** Outsourcing***

FaceSwap *2021*

Honors & Awards

- Silver Reviewer for ICML 2026
- Graduation Project 2021 *(2nd)* Project: Lip Reading
- MAIC Surgical Instruments Segmentation 2021 *(3rd)* Project: Medical Segmentation
- Korea SW Central University Hackton 2021 *(3rd)* Project: Health Care Body Position
- SM AI Competition 2020 *(2nd & 4rd)* Project: Image Inpainting & Music Recommendation

Certifications

- Tensorflow Developer Certificate 2022
- YONSEI Medical Scientist Training Program 2023
- Summer Program in University of Southern California 2023
- Best Academic Paper Award at Yonsei University